

ERP HRMS Technical Implementation & Infrastructure Configuration Guide

Workforce Management | Succession Planning | Recruitment | Onboarding
Integration

Build-ready documentation for Dev, Infra, Security, HRIS and Project Teams

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1. How to Use This Document

This document converts the functional ERP HR design into a build-ready technical guide. It is written for two teams: the development team that will implement the modules and the infrastructure team that will configure the platform, environments, security, CI/CD, data stores, observability and operational controls.

Implementation stance

This guide is product-agnostic but borrows enterprise HR patterns from SAP-style HCM systems: core HR as the system of record, position management as the control point, recruiting and onboarding as the hire-to-employee flow, workforce management as the operational layer, and succession planning as the talent continuity layer.

Team	What to focus on
Developers	Domain model, API contracts, workflow states, maker-checker rules, module service boundaries, event payloads, validation rules, tests and integration patterns.
Infrastructure team	Kubernetes/OpenShift topology, namespaces, ingress, TLS, database, Redis, object storage, IAM, secrets, backup, monitoring, CI/CD runners and release strategy.
Security team	RBAC, segregation of duties, audit logs, data classification, encryption, approval controls, privileged access, retention and compliance.
HRIS / Business Analysts	Process configuration, approval matrices, fields, master data ownership, reporting, user roles, UAT scenarios and change management.
Project team	Milestones, dependencies, delivery phases, go-live checklist and future roadmap.

Assumption	Decision
Architecture style	Start as a modular ERP platform with clear service boundaries. It can later split into microservices if scale and team structure justify it.
Hosting model	Bank-grade on-prem, private cloud or sovereign cloud deployment using Kubernetes/OpenShift. Public cloud is optional, not mandatory.
Identity provider	Integrate with enterprise AD / Azure AD / LDAP through Keycloak, OIDC or SAML.
Database	PostgreSQL as the transactional database, with read replicas for reporting where needed.
Cache and jobs	Redis for sessions, cache, Celery/RQ queues and short-lived locks.
Eventing	Kafka or a compatible event bus for future-ready integration and analytics pipelines.
Documents	Object storage such as MinIO/S3-compatible storage for CVs, offer letters, contracts and onboarding files.
Security posture	Zero-trust access, least privilege, immutable audit, encryption in transit and at rest.

2. Architecture Principles and Enterprise HR Reference Model

Enterprise HR systems commonly centralize core HR processes, talent processes, workforce planning, recruiting, onboarding, time and attendance, and analytics into a connected HCM suite. SAP describes SuccessFactors HCM as a connected suite supporting core HR and payroll, talent management, HR analytics, workforce planning and employee experience. SAP also describes HRMS capabilities as including employee data storage, benefits administration, time and attendance, payroll, recruiting, onboarding, performance, learning, compensation and succession planning. See References [1].

This implementation uses the same high-level separation: Core HR owns employee and organization master data; Recruitment owns the candidate pipeline; Onboarding converts accepted candidates to employees; Workforce Management owns assignments, attendance, rosters and manpower; Succession Planning owns critical roles, successor readiness and development plans.

Principle	Technical meaning	Design implication
Position-first HR	The position is the anchor object between workforce planning, recruitment and succession.	Recruitment requisitions must reference approved vacant positions. Succession plans must reference critical positions.

Core HR as source of truth	Employee, employment and organization records are not duplicated across modules.	Other modules consume employee and position IDs through APIs/events.
Workflow by configuration	Approvals should be rule-driven instead of hardcoded.	Approval matrix table + workflow engine + role/grade/cost-center rules.
Maker-checker by default	Sensitive HR transactions require independent review.	Maker cannot approve own transaction; all approvals are audited.
Effective dating	Historical changes must be traceable with valid_from and valid_to.	Employee assignment, position, grade and organization changes use effective dates.
Event-driven integration	Modules publish domain events after approved state changes.	Analytics, notification, IAM, payroll and reporting subscribe to events.
Future-ready data layer	Operational data should be analytics-ready without overloading OLTP.	Use outbox pattern + Kafka + data warehouse/lakehouse later.
Secure by design	PII and confidential talent data require strict access.	RBAC/ABAC, encryption, audit, masking, and data retention rules.

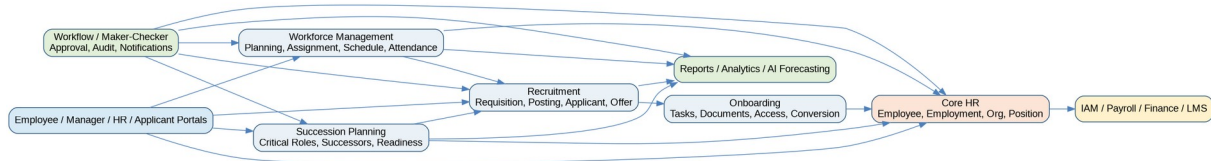


Figure 1 - Enterprise HR reference model used by this ERP HRMS.

3. Recommended Technology Stack

The stack below is intentionally future-ready but still practical for a bank or enterprise IT team. It supports an initial MVP, controlled production hardening and later expansion into event-driven analytics, AI workforce planning and mobile-first HR services.

Layer	Recommended technology	Why
Frontend	React, Vue or Angular; TypeScript preferred	Enterprise-grade UI, reusable components, strong typing and role-based screens.
Applicant portal	Separate public frontend or isolated route/app	Public-facing recruitment portal should be isolated from internal HR screens.
Backend API	Django REST Framework / FastAPI / Spring Boot / .NET	All are enterprise-ready. If your team uses Python, Django + DRF is fastest for ERP CRUD and workflow screens.
Workflow jobs	Celery/RQ with Redis or Temporal/Camunda for advanced orchestration	Celery is simple for MVP; Temporal/Camunda is stronger for long-running enterprise workflows.
Database	PostgreSQL 15+	Strong relational integrity, JSONB where needed, row-level constraints and mature HA options.
Cache/session	Redis	Session storage, cache, rate limiting, locks and job queues.
Event bus	Kafka / Redpanda / RabbitMQ	Kafka is preferred for future analytics and event replay; RabbitMQ is simpler for commands/tasks.
Object storage	MinIO/S3-compatible storage	CVs, documents, offer letters, ID copies, policy docs and generated PDFs.
IAM/SSO	Keycloak, AD, Azure AD, LDAP, OIDC/SAML	Central authentication, MFA, SSO and role mapping.
Secrets	HashiCorp Vault or Kubernetes sealed-secrets	Secure secret distribution and rotation.
Containers	Docker/Podman	Repeatable builds and local dev parity.
Orchestration	Kubernetes / OpenShift	Scaling, self-healing, ingress, secret/config management and enterprise deployment controls.
Observability	Prometheus, Grafana, Loki/ELK/OpenSearch, OpenTelemetry	Metrics, logs, traces, dashboards and alerts.
CI/CD	GitLab CI, Azure DevOps, GitHub Actions or Jenkins	Pipeline-driven build, test, security scan and deployment.

Recommended implementation approach

Use a modular monolith for v1: one deployable backend with separate apps/modules, one database with clear schemas, one workflow engine and one event outbox. Split into microservices only after domain boundaries, traffic and team ownership are stable.

Stage	Architecture style	When to use
MVP / Phase 1	Modular monolith	Fastest delivery, fewer moving parts, easier transactions and simpler operations.
Production Phase 2	Modular monolith + event bus	Still simple, but analytics, notifications and integration become asynchronous.
Scale Phase 3	Selected microservices	Split high-change or high-load domains such as recruitment portal, workflow, reporting and document services.
Enterprise Phase 4	Domain platform + data products	Add lakehouse, skill graph, AI recommendations, mobile apps and API ecosystem.

4. End-to-End Solution Architecture

The ERP HRMS is split into presentation, API, domain service, workflow, data, integration and platform layers. Each layer has a clear responsibility and deployment boundary.

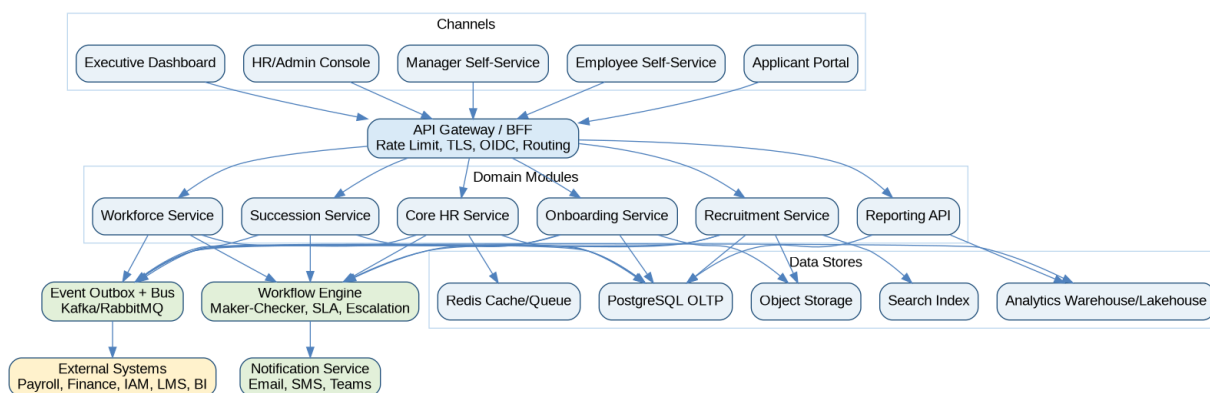


Figure 2 - Logical application architecture with domain modules and shared platform services.

Component	Responsibility	Owned data
API Gateway / BFF	Entry point for web/mobile clients, route requests, validate tokens, rate limit, API versioning.	No business data. Request logs only.
Core HR Service	Employee, person, employment, organization, position, job, grade and assignment master data.	core_* tables.
Recruitment Service	Job requisition, posting, applicant, application, shortlist, interview, offer.	rec_* tables.
Onboarding Service	Pre-boarding tasks, documents, access requests, asset requests, conversion checklist.	onb_* tables.
Workforce Service	Workforce plan, position requests, attendance, roster, overtime, transfer, promotion.	wfm_* tables.
Succession Service	Critical role register, talent profile, successor nomination, readiness, development plans.	succ_* tables.
Workflow Engine	Reusable maker-checker lifecycle, approval routing, delegation, escalation and audit.	workflow_* tables.
Notification Service	Send approval, SLA, onboarding, interview and exception notifications.	notification_* tables.
Reporting API	Read-optimized reporting endpoints and dashboards.	Views/materialized views/read replicas.
Integration Adapter	Payroll, IAM, Finance, LMS and external job boards.	integration logs, mapping tables.

5. Application Modules and Service Boundaries

Module	Primary capabilities	Must not own
Core HR	Employee master, organization, job, position, grade, employment lifecycle, manager line.	Candidate pipeline, attendance transactions, succession readiness.
Recruitment	Requisition, posting, applicant profile, application, screening, interview, offer.	Employee master creation after approval; it sends to onboarding/Core HR.
Onboarding	New hire checklist, document capture, access/asset tasks, pre-employment activities.	Recruitment selection logic or long-term employee profile.
Workforce Management	Workforce plan, headcount, attendance, schedules, overtime, transfer, promotion.	Applicant records and confidential succession rankings.
Succession Planning	Critical roles, successors, readiness, talent pools, risk and development plans.	Recruitment pipeline; it triggers requisitions but does not run hiring.
Workflow	Approval engine, status changes, delegation, SLA, comments, audit.	Domain-specific validation; domain services own validation.
Reporting	Aggregated dashboards, KPI views, exports.	Transactional write operations.

Recommended repository structure

```

erp-hrms/
├── apps/
│   ├── core_hr/      # employee, org, job, position, assignment
│   ├── recruitment/  # requisition, posting, applicant, pipeline, offer
│   ├── onboarding/   # checklist, documents, asset/access tasks
│   ├── workforce/    # workforce plan, roster, attendance, overtime
│   ├── succession/   # critical roles, successors, readiness, development
│   ├── workflow/     # maker-checker engine, approval matrix, comments
│   ├── notification/ # email/SMS/Teams notifications
│   ├── integration/  # payroll, IAM, finance, LMS, job boards
│   ├── reporting/    # KPI APIs, materialized views, dashboards
│   └── audit/        # immutable audit log and history
├── config/
│   ├── settings/     # dev, test, uat, prod settings
│   ├── urls.py
│   └── asgi.py/wsgi.py
├── frontend/
│   ├── applicant-portal/
│   ├── employee-portal/
│   └── admin-console/
├── infra/
│   ├── docker-compose.yml
│   ├── helm/
│   ├── k8s/base/
│   ├── k8s/overlays/dev/
│   ├── k8s/overlays/uat/
│   └── k8s/overlays/prod/
├── docs/
│   ├── architecture/
│   ├── api/
│   ├── runbooks/
│   └── security/
├── tests/
│   ├── unit/
│   ├── integration/
│   ├── e2e/
│   └── performance/

```

6. Recruitment Module - Technical Design

The recruitment module is the controlled entry point for external and internal candidates. It should only create a hiring request after an approved position or approved strategic hiring need exists. SAP SuccessFactors documentation also

highlights the integration between Recruiting Management and Onboarding as part of the hiring process. See References [5].

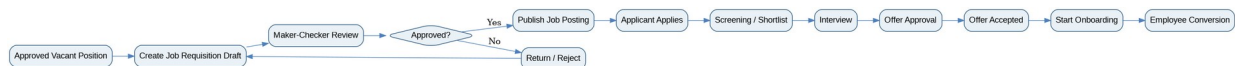


Figure 3 - Recruitment module process from approved position to employee conversion.

Sub-module	Features	Key controls
Job Requisition	Create requisition from approved position, specify hiring reason, grade, location, reporting manager, headcount, budget and job description.	Position must be approved and vacant; budget and grade validation required.
Job Posting	Internal, external or confidential posting; posting date, closing date, channels, eligibility and screening questions.	Maker-checker required before publication.
Applicant Portal	Registration, login, profile, CV upload, application status, document submission.	Public DMZ isolation, rate limits, CAPTCHA, malware scan for uploads.
Candidate Pipeline	Longlist, shortlist, HR screening, hiring manager review, interview stages, scoring, comments.	Panel visibility and conflict-of-interest declaration.
Interview Management	Schedule, panel, location/Teams link, scoring form and feedback.	Feedback locked after submission. Audit all score changes.
Offer Management	Offer draft, compensation validation, approval, candidate acceptance, expiry, negotiation notes.	HR + Finance checker for grade/pay/budget.
Onboarding Handoff	Send candidate, offer, position and required task templates to onboarding module.	Only after accepted offer and required approvals.

Recruitment status	Meaning	Allowed next status
DRAFT	Requisition created by maker.	SUBMITTED, CANCELLED
SUBMITTED	Pending checker review.	RETURNED, REJECTED, CHECKED
CHECKED	Verified by checker.	APPROVED, REJECTED
APPROVED	Ready for posting.	POSTED, CANCELLED
POSTED	Visible to candidates.	CLOSED, SUSPENDED
SCREENING	Candidates being reviewed.	INTERVIEW, REJECTED
INTERVIEW	Candidate in interview stages.	OFFER_DRAFT, REJECTED
OFFER_DRAFT	Offer being prepared.	OFFER_SUBMITTED
OFFER_APPROVED	Offer approved for candidate.	OFFER_SENT
OFFER_ACCEPTED	Candidate accepted.	ONBOARDING_STARTED
ONBOARDING_STARTED	Onboarding module owns transition.	EMPLOYEE_CREATED

Validation rule	Implementation detail
Requisition must reference position_id	Foreign key to core_position. If strategic hiring without position is required, create a position request first.
Posting requires approved requisition	API rejects POST /job-postings unless requisition.status == APPROVED.
Candidate cannot apply twice to same posting	Unique constraint: applicant_id + job_posting_id.
Offer requires candidate stage eligibility	Only shortlisted/interview-passed candidates can receive offer draft.
Offer requires budget validation	Call Finance adapter or compare pay band, grade and approved budget.
Onboarding starts once offer is accepted	Publish candidate.offer_accepted event and create onboarding case.

7. Workforce Management Module - Technical Design

The workforce module operationalizes employee assignment, attendance, roster, overtime and manpower planning. SAP describes workforce management capabilities around time tracking, scheduling, attendance and time management. The ERP implementation here extends those operational capabilities with position control and workforce gap analysis. See References [4].

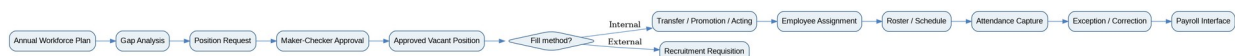


Figure 4 - Workforce management process and integration to recruitment/payroll.

Capability	Technical implementation	Maker-checker required
Workforce plan	Yearly/quarterly plan per department, cost center, job family and grade.	Yes, HR + Finance approval.
Position request	Draft position from manager/HR with justification, grade, budget and location.	Yes.
Employee assignment	Effective-dated assignment to position, department, manager and cost center.	Yes for transfer/promotion; no for system-created onboarding assignment after approval.
Schedule/roster	Shift template, roster calendar, publication workflow and employee visibility.	Yes for published rosters in regulated operations.
Attendance	Clock-in/out from QR, NFC, biometric, web/mobile or imported device logs.	No for raw logs; yes for manual correction.
Attendance exception	Late, absent, early out, missing clock, mismatch with leave/schedule.	Yes for correction or override.
Overtime	Request, eligibility check, approval, actual hours matching and payroll export.	Yes.
Transfer/promotion	Move employee between position, department, branch, manager or grade.	Yes, with releasing and receiving approvals.

Workforce event	Trigger	System action
position.approved	Position request approved.	Create vacant position and update headcount dashboard.
position.vacant	Resignation, retirement, transfer, promotion or termination.	Check critical role and internal successors; optionally trigger recruitment.
employee.assigned	Employee assigned to position.	Set position status to occupied and update manager/org chart.
attendance.exception_created	Attendance rules detect issue.	Notify employee/manager; create correction workflow if submitted.
overtime.approved	Overtime request approved.	Export to payroll after attendance matching.
workforce_gap.detected	Plan vs actual mismatch.	Create HR action: recruit, redeploy, overtime or freeze.

8. Succession Planning Module - Technical Design

Succession planning protects critical roles by identifying ready successors, talent gaps and development actions. SAP describes succession-related AI insights that help planners understand role alignment, performance, achievements, strengths and development areas directly from succession views. See References [3].



Figure 5 - Succession planning flow from critical role to development/recruitment action.

Capability	Technical implementation	Security classification
Critical role register	Position-level flag with criticality reason, business impact, regulatory impact, recovery time objective and owner.	Confidential HR.
Talent profile	Skills, certifications, employment history, performance summary, potential rating, mobility, career interests.	Confidential; partial employee view only.
Successor nomination	Nominee, target role, readiness, evidence, risk, development gaps and approver comments.	Highly confidential.
Readiness scoring	Ready now, <1 year, 1-2 years, 3+ years, emergency-only, no successor.	Highly confidential.
9-box grid	Performance vs potential matrix with calibration history.	Restricted to HR/talent committee.
Development plan	Learning, mentoring, certifications, acting role, project exposure and due dates.	Manager/employee view depends on policy.
Succession risk	Weighted score combining vacancy risk, no successor, readiness gap, retirement risk, attrition risk.	Executive/HR dashboard.

Risk factor	Example scoring rule	Purpose
No successor	+40 points	Immediate red flag for critical role.
Only one successor	+10 points	Weak bench strength.
Successor not ready within 2 years	+15 points	Development gap.
Incumbent retirement eligible	+20 points	Continuity risk.
High attrition risk	+20 points	Flight risk signal from HR analytics.
Regulatory-critical role	+15 points	Business continuity and compliance risk.
No development plan	+5 points	Governance gap.

9. Maker-Checker Workflow Engine

The workflow engine is shared across recruitment, workforce management, succession planning, onboarding and core HR. It prevents each module from building a separate approval mechanism. The workflow engine stores state, approvers, comments, SLA, delegation, escalation and audit events.

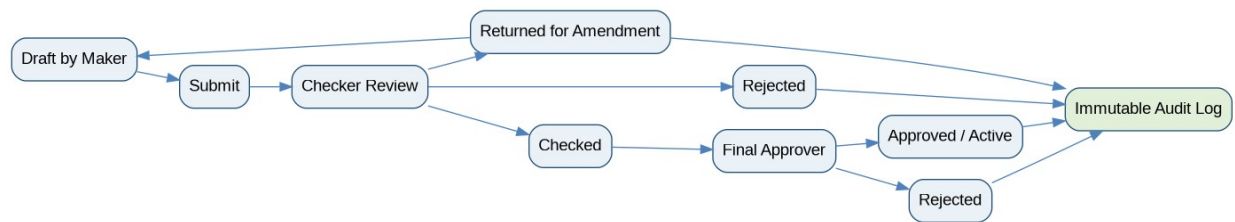


Figure 6 - Generic maker-checker workflow lifecycle.

Workflow object	Purpose
workflow_request	Header record for any approval transaction. Stores module, object type, object id, maker, status and SLA.
workflow_step	Individual approval step such as HR checker, finance checker, department head or talent committee.
workflow_actor	Assigned approvers, delegates, watchers and escalation owners.
workflow_comment	Approval/rejection/return notes.
workflow_attachment	Supporting documents for approvals.
workflow_history	Append-only state transition trail.
audit_log	Immutable business audit across all modules.

Example workflow rule configuration

```

{
  "workflow_code": "REC_JOB_POSTING_APPROVAL",
  "applies_to": "recruitment.job_posting",
  "conditions": {
    "grade_level": ["G5", "G6", "G7"],
    "posting_visibility": ["EXTERNAL", "BOTH"]
  },
  "steps": [
    {"step": 1, "role": "RECRUITMENT_CHECKER", "sla_hours": 24},
    {"step": 2, "role": "HR_MANAGER", "sla_hours": 24},
    {"step": 3, "role": "FINANCE_CHECKER", "required_when": "budget_impact == true"}
  ],
  "segregation_of_duties": true,
  "maker_cannot_approve": true,
  "audit_required": true
}

```

Control	Implementation requirement
Maker cannot approve own transaction	Workflow engine checks maker_user_id != approver_user_id.
Checker must provide reason on reject/return	reject_reason required for status RETURNED or REJECTED.
Approval matrix versioning	Store workflow_rule.version and effective dates. Old transactions keep old rule version.
Escalation	If SLA breached, notify escalation owner; do not auto-approve unless

requisition_id	UUID PK	Yes	Hiring request ID.
position_id	UUID FK	Yes	Approved vacant position.
hiring_reason	enum	Yes	replacement, expansion, succession_gap.
requested_by	UUID FK	Yes	Manager/HR maker.
status	enum	Yes	draft to approved.
target_start_date	date	No	Expected hire date.
workflow_request_id	UUID FK	No	Approval linkage.

10.x Data Dictionary - rec_job_posting

Field	Type	Required	Notes
job_posting_id	UUID PK	Yes	Public/internal posting.
requisition_id	UUID FK	Yes	Approved requisition.
title	varchar(255)	Yes	Published title.
visibility	enum	Yes	internal, external, both, confidential.
opening_date/closing_date	date	Yes	Posting window.
status	enum	Yes	draft, posted, suspended, closed.

10.x Data Dictionary - rec_applicant

Field	Type	Required	Notes
applicant_id	UUID PK	Yes	Candidate identity.
email	varchar(255)	Yes	Unique login email.
full_name	varchar(255)	Yes	Candidate name.
profile_status	enum	Yes	active, locked, withdrawn.
consent_version	varchar(50)	Yes	Privacy consent version.

10.x Data Dictionary - rec_application

Field	Type	Required	Notes
application_id	UUID PK	Yes	Application to a job posting.
job_posting_id	UUID FK	Yes	Target posting.
applicant_id	UUID FK	Yes	Applicant.
stage	enum	Yes	applied, screened, interview, offer, hired, rejected.
score	numeric	No	Screening/interview score.

10.x Data Dictionary - rec_offer

Field	Type	Required	Notes
offer_id	UUID PK	Yes	Offer record.
application_id	UUID FK	Yes	Selected application.
position_id	UUID FK	Yes	Position to fill.
grade_id	UUID FK	Yes	Approved grade.
status	enum	Yes	draft, submitted, approved, sent, accepted, declined.
accepted_at	timestamp	No	Candidate acceptance time.

10.x Data Dictionary - onb_case

Field	Type	Required	Notes
onboarding_id	UUID PK	Yes	Onboarding case.
offer_id	UUID FK	Yes	Accepted offer.
candidate_person_id	UUID FK	No	Pre-hire identity.
status	enum	Yes	started, in_progress, completed, waived, cancelled.
target_start_date	date	Yes	Employee start date.

10.x Data Dictionary - wfm_workforce_plan

Field	Type	Required	Notes
plan_id	UUID PK	Yes	Workforce plan header.
org_unit_id	UUID FK	Yes	Department/unit.
planning_year	int	Yes	Budget/planning year.
required_headcount	int	Yes	Planned headcount.
approved_headcount	int	No	After approval.
status	enum	Yes	draft, submitted, approved.

10.x Data Dictionary - wfm_attendance_log

Field	Type	Required	Notes
attendance_id	UUID PK	Yes	Attendance event.
employee_id	UUID FK	Yes	Employee.
work_date	date	Yes	Attendance date.
clock_in/clock_out	timestamp	No	Captured times.
source	enum	Yes	mobile, biometric, web, import, manual.
status	enum	Yes	normal, late, absent, exception.

10.x Data Dictionary - wfm_overtime_request

Field	Type	Required	Notes
overtime_id	UUID PK	Yes	Overtime request.
employee_id	UUID FK	Yes	Employee.
requested_hours	numeric	Yes	Requested OT.
approved_hours	numeric	No	Approved OT.
status	enum	Yes	draft, submitted, approved, rejected, paid.

10.x Data Dictionary - succ_succession_plan

Field	Type	Required	Notes
plan_id	UUID PK	Yes	Succession plan.
position_id	UUID FK	Yes	Critical position.
criticality_type	enum	Yes	business, leadership, regulatory, technical.
risk_score	int	Yes	Calculated risk score.
status	enum	Yes	draft, submitted, approved, active, archived.

10.x Data Dictionary - succ_successor_nomination

Field	Type	Required	Notes
nomination_id	UUID PK	Yes	Nomination record.
plan_id	UUID FK	Yes	Succession plan.
employee_id	UUID FK	Yes	Nominated employee.
readiness	enum	Yes	ready_now, lt_1_year, 1_2_years, 3_plus_years.
evidence_summary	text	Yes	Required evidence.
status	enum	Yes	draft, submitted, approved, rejected.

10.x Data Dictionary - workflow_request

Field	Type	Required	Notes
workflow_request_id	UUID PK	Yes	Approval request header.
module_code	varchar(50)	Yes	rec, wfm, succ, core, onb.
object_type	varchar(100)	Yes	e.g. job_posting.
object_id	UUID	Yes	Domain object ID.

maker_user_id	UUID FK	Yes	Creator.
status	enum	Yes	draft/submitted/approved/etc.

10.x Data Dictionary - audit_log

Field	Type	Required	Notes
audit_id	UUID PK	Yes	Immutable audit ID.
actor_user_id	UUID FK	Yes	User/system actor.
action	varchar(100)	Yes	create, update, approve, reject.
object_type/object_id	varchar/UUID	Yes	Target object.
before_json/after_json	jsonb	No	Change snapshot.
created_at	timestamp	Yes	Immutable timestamp.

Example SQL constraints and indexes

```
-- Critical constraints for ERP integrity
ALTER TABLE rec_job_requisition
ADD CONSTRAINT fk_requisition_position
FOREIGN KEY (position_id) REFERENCES core_position(position_id);

ALTER TABLE rec_application
ADD CONSTRAINT uq_applicant_per_posting
UNIQUE (applicant_id, job_posting_id);

ALTER TABLE core_employee_assignment
ADD CONSTRAINT uq_primary_assignment_per_employee
UNIQUE (employee_id, is_primary) DEFERRABLE INITIALLY DEFERRED;

-- Recommended indexes
CREATE INDEX idx_position_status_org ON core_position(status, org_unit_id);
CREATE INDEX idx_requisition_position ON rec_job_requisition(position_id);
CREATE INDEX idx_attendance_employee_date ON wfm_attendance_log(employee_id, work_date);
CREATE INDEX idx_successor_position ON succ_succession_plan(position_id, status);
CREATE INDEX idx_audit_object ON audit_log(object_type, object_id, created_at DESC);
```

11. API Design and Integration Contracts

Use REST for user-driven CRUD and command operations, and events for asynchronous integration. API endpoints should be versioned from day one: /api/v1/.... Internal service-to-service calls should use short-lived tokens and service identities.

Method	Endpoint	Purpose	Allowed role
POST	/api/v1/positions/requests	Create position request draft	Manager/HR Maker
POST	/api/v1/positions/{id}/submit	Submit position request to workflow	Maker
POST	/api/v1/workflows/{id}/approve	Approve workflow step	Checker/Approver
POST	/api/v1/recruitment/requisitions	Create requisition from approved position	Recruitment Maker
POST	/api/v1/recruitment/postings/{id}/publish	Publish approved job posting	Recruitment Checker/HR
POST	/api/v1/applicant/register	Applicant registration	Public
POST	/api/v1/applicant/applications	Apply to job posting	Applicant
POST	/api/v1/recruitment/applications/{id}/shortlist	Shortlist candidate	Recruiter/Manager
POST	/api/v1/recruitment/offers	Create offer draft	Recruiter
POST	/api/v1/recruitment/offers/{id}/accept	Candidate accepts offer	Applicant
POST	/api/v1/onboarding/cases	Create onboarding case	System/HR
POST	/api/v1/onboarding/{id}/complete	Complete onboarding and trigger employee draft	HR/Onboarding
POST	/api/v1/core/employees/convert-from-onboarding	Create employee from onboarding	HR Maker

POST	/api/v1/workforce/attendance/corrections	Request attendance correction	Employee/Manager/HR
POST	/api/v1/workforce/overtime	Request overtime	Employee/Manager
POST	/api/v1/succession/critical-roles	Create critical role draft	HR Talent Maker
POST	/api/v1/succession/nominations	Nominate successor	Manager/HR Talent
GET	/api/v1/reports/workforce/headcount	Headcount dashboard data	HR/Executive
GET	/api/v1/reports/succession/risk	Succession risk dashboard data	HR/Executive

Example recruitment requisition API

POST /api/v1/recruitment/requisitions

Authorization: Bearer <token>

Content-Type: application/json

```
{
  "position_id": "0d2b0af4-0c97-4a8d-9f55-0fded0a11111",
  "hiring_reason": "REPLACEMENT",
  "target_start_date": "2026-08-01",
  "justification": "Replacement for approved vacant Head of IT Security position.",
  "requested_by_employee_id": "EMP-000102",
  "budget_confirmed": false
}
```

Response 201

```
{
  "requisition_id": "REQ-2026-00018",
  "status": "DRAFT",
  "workflow_required": true,
  "next_action": "SUBMIT_FOR_APPROVAL"
}
```

Example workflow approval API

POST /api/v1/workflows/{workflow_request_id}/approve

Authorization: Bearer <token>

Content-Type: application/json

```
{
  "decision": "APPROVE",
  "comment": "Position, grade and budget have been verified.",
  "step_id": "HR_CHECKER_STEP_1"
}
```

Response 200

```
{
  "workflow_request_id": "WF-2026-001221",
  "status": "CHECKED",
  "next_step": "FINANCE_CHECKER",
  "audit_id": "AUD-2026-99881"
}
```

12. Event-Driven Integration and Data Flows

Use an outbox table to guarantee event publication after database commit. Each module writes its transaction and outbox event in the same database transaction. A background publisher sends outbox records to Kafka/RabbitMQ and marks them as published. This avoids lost events when the app crashes after commit but before publish.

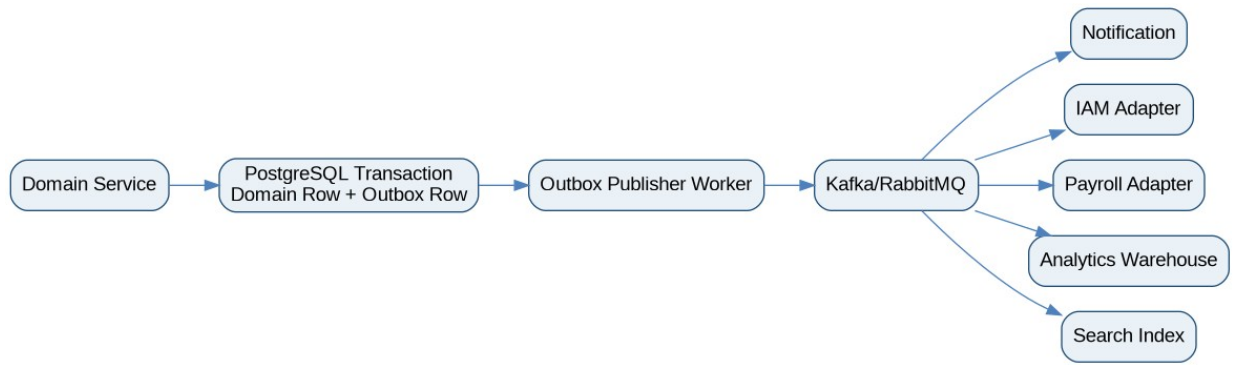


Figure 8 - Reliable event outbox pattern for integration and analytics.

Event name	Publisher	Subscribers	Payload minimum
position.approved	Workforce/Core HR	Recruitment, Reporting	position_id, org_unit_id, job_id, grade, cost_center, approved_at
position.vacant	Core HR	Recruitment, Succession, Reporting	position_id, incumbent_employee_id, vacancy_reason, effective_date
requisition.approved	Recruitment	Job Posting, Reporting	requisition_id, position_id, approved_by, approved_at
job_posting.published	Recruitment	Applicant Portal, Search, Reporting	posting_id, title, visibility, closing_date
candidate.offer_accepted	Recruitment	Onboarding, Reporting	offer_id, applicant_id, position_id, start_date
onboarding.completed	Onboarding	Core HR, IAM, Payroll, Reporting	onboarding_id, offer_id, completed_at
employee.created	Core HR	Workforce, IAM, Payroll, Reporting	employee_id, position_id, start_date, manager_id
attendance.approved	Workforce	Payroll, Reporting	employee_id, work_date, normal_hours, overtime_hours
successor.not_found	Succession	Recruitment, HR Dashboard	position_id, risk_score, target_fill_date
development_plan.created	Succession	LMS, Manager Portal	employee_id, plan_id, skill_gap, due_date

Example domain event payload

```

{
  "event_id": "evt_01HXWFM6Y8",
  "event_name": "candidate.offer_accepted",
  "event_version": 1,
  "occurred_at": "2026-05-23T09:00:00+08:00",
  "correlation_id": "corr_REQ-2026-00018",
  "source": "recruitment-service",
  "data": {
    "offer_id": "OFF-2026-00088",
    "application_id": "APP-2026-12345",
    "applicant_id": "APL-2026-00991",
    "position_id": "POS-IT-SEC-001",
    "accepted_start_date": "2026-08-01",
    "grade_id": "G7",
    "onboarding_template_code": "BANK_STANDARD_NEW_HIRE"
  }
}

```

13. Security, RBAC, SoD and Audit Controls

Area	Requirement
Authentication	OIDC/SAML SSO for internal users; secure applicant login with MFA

	optional; service accounts for integrations.
Authorization	RBAC for coarse access; ABAC for sensitive cases such as department, grade, own-record, manager-of and confidentiality.
Segregation of duties	Maker cannot approve own transaction. Recruiter cannot approve own offer. HR maker cannot activate employee without checker.
PII protection	Encrypt sensitive fields where applicable; restrict export; mask data in lower environments.
Document security	Object storage private buckets, pre-signed URLs, malware scan, file type allowlist and retention rules.
Audit logging	Append-only audit for create/update/approve/reject/login/export/admin actions.
Session security	Short-lived access tokens, refresh token rotation, idle timeout, secure cookies, CSRF protection.
Transport security	TLS 1.2+ everywhere, internal mTLS optional for service mesh.
Secrets	No secrets in Git; use Vault, external secrets operator or sealed-secrets.
Admin access	Privileged access management, break-glass account, audit and periodic access review.

Role	Recruitment	Workforce	Succession	Core HR	Workflow
Applicant	Own applications only	No access	No access	No access	No access
Employee	Internal postings if allowed	Own schedule/attendance	Own development plan if policy allows	Own profile read	Submit own requests
Manager	Review candidates for own requisitions	Team attendance/roster	Nominate successors in own org	Team read subset	Approve team workflows
Recruiter	Manage requisitions/candidates/offers	Read position vacancy	No confidential succession by default	Read approved position data	Submit/approve based on role
HR Maker	Create/update HR records	Create workforce transactions	Create succession drafts	Create employee drafts	Submit but not self-approve
HR Checker	Review recruitment HR steps	Review corrections/movements	Review nominations/readiness	Approve employee activation	Approve assigned steps
Finance Checker	Offer/budget review	Budget/headcount review	External hiring budget review	Cost center review	Approve finance steps
Talent Committee	Read strategic requisitions	Read workforce risk	Approve succession plans	Limited read	Approve committee steps
System Admin	Config only, no HR business approval by default	Config only	Config only	Config only	Maintain workflow roles

Audit event	Required fields
LOGIN_SUCCESS / LOGIN_FAILURE	user_id, timestamp, ip_address, user_agent, auth_method.
RECORD_CREATED / RECORD_UPDATED	object_type, object_id, before_json, after_json, actor, timestamp.
WORKFLOW_APPROVED / REJECTED / RETURNED	workflow_request_id, step_id, decision, comment, actor, timestamp.
EXPORT_PERFORMED	report_name, filters, row_count, actor, timestamp.
DOCUMENT_DOWNLOADED	document_id, object_type, actor, timestamp, ip_address.
ROLE_CHANGED	target_user, old_roles, new_roles, changed_by, approval_ref.
BREAK_GLASS_USED	actor, reason, duration, reviewed_by, review_date.

14. Infrastructure Deployment Architecture

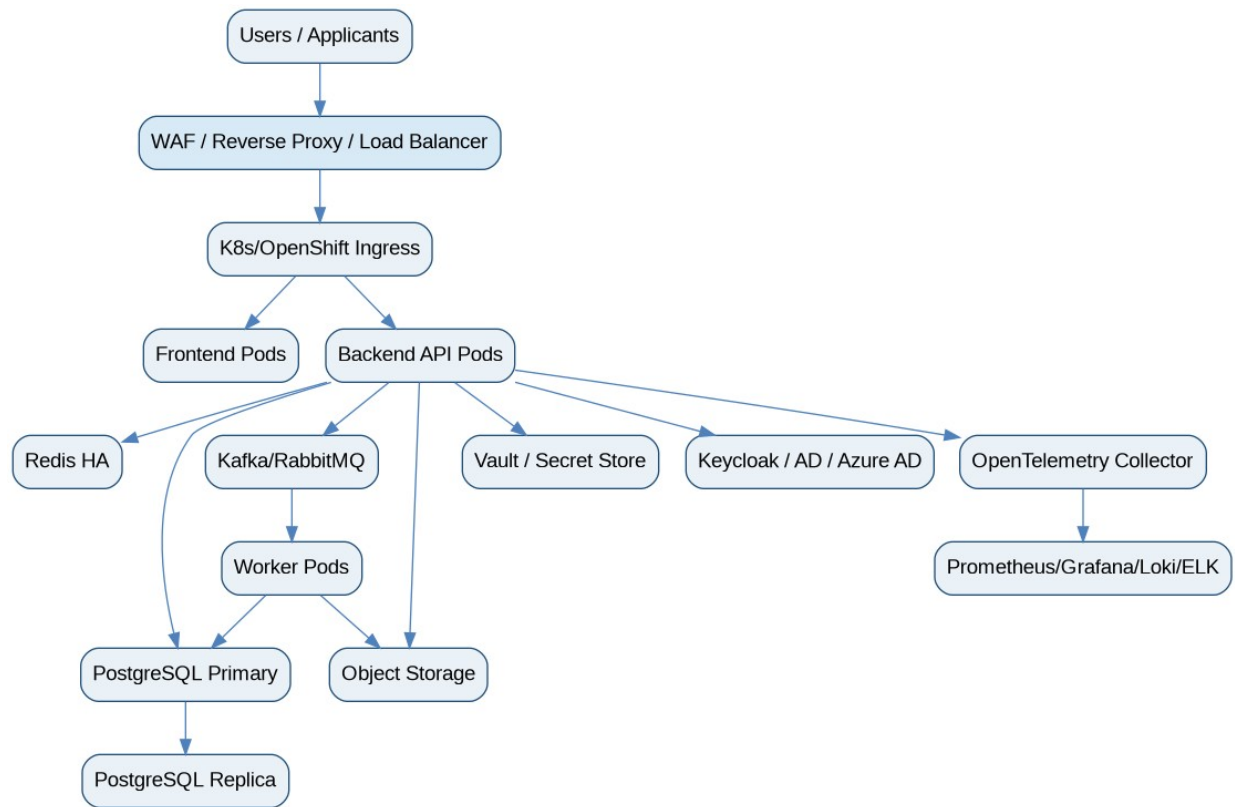


Figure 9 - Recommended production deployment topology.

Environment	Purpose	Minimum recommendation
DEV	Developer integration and feature testing.	Single namespace, small Postgres/Redis, mock integrations, sample data.
SIT	System integration test with shared services.	Separate namespace, test SSO, integration stubs or non-prod endpoints.
UAT	Business user testing and workflow validation.	Production-like topology, masked data, test IAM roles.
PREPROD	Release rehearsal, performance, security scans and cutover testing.	Same as production scale down; production configs except data/endpoints.
PROD	Live enterprise HRMS.	HA database, HA ingress, replicas, monitoring, backup, DR and strict access.

Component	DEV	UAT/PREPROD	PROD baseline
Frontend pods	1	2	2-4 with HPA
Backend API pods	1-2	2	3+ with HPA
Worker pods	1	2	2+ by queue type
PostgreSQL	Single instance	Single/replica	Primary + replica + backups
Redis	Single instance	Single/HA	Sentinel/Cluster/managed HA
Object storage	Local MinIO	MinIO HA or shared storage	MinIO HA/S3 enterprise storage
Ingress	Single controller	HA controller	HA + WAF/load balancer
Monitoring	Basic	Prometheus/Grafana	Full metrics/logs/traces/alerts
Backup	Manual/daily	Scheduled	Scheduled + offsite + restore test

15. Kubernetes/OpenShift Configuration Guide

Namespace	Purpose	Notes
hrms-dev	Development deployment.	Lower resource limits, debug logs allowed.
hrms-uat	UAT deployment.	Production-like role mapping and masked data.
hrms-prod-blue	Current or standby production blue	Used for blue-green release.

	environment.	
hrms-prod-green	Current or standby production green environment.	Used for blue-green release.
hrms-observability	Monitoring/logging stack.	Shared Prometheus/Grafana/Loki/OpenSearch.
hrms-platform	Shared platform services.	Ingress, cert-manager, external-secrets, service mesh if used.

Example namespace and quota

```

apiVersion: v1
kind: Namespace
metadata:
  name: hrms-prod-blue
  labels:
    app.kubernetes.io/part-of: erp-hrms
    environment: prod
    color: blue
---
apiVersion: v1
kind: ResourceQuota
metadata:
  name: hrms-prod-quota
  namespace: hrms-prod-blue
spec:
  hard:
    requests.cpu: "8"
    requests.memory: 16Gi
    limits.cpu: "16"
    limits.memory: 32Gi
    pods: "80"

```

Example API deployment

```

apiVersion: apps/v1
kind: Deployment
metadata:
  name: hrms-api
  namespace: hrms-prod-blue
spec:
  replicas: 3
  selector:
    matchLabels:
      app: hrms-api
      color: blue
  template:
    metadata:
      labels:
        app: hrms-api
        color: blue
    spec:
      containers:
        - name: api
          image: registry.internal/erp-hrms/api:1.0.0
          ports:
            - containerPort: 8000
          envFrom:
            - configMapRef:
                name: hrms-api-config
            - secretRef:
                name: hrms-api-secret
          readinessProbe:

```

```
httpGet:
  path: /health/ready
  port: 8000
initialDelaySeconds: 20
periodSeconds: 10
livenessProbe:
  httpGet:
    path: /health/live
    port: 8000
  initialDelaySeconds: 40
  periodSeconds: 20
resources:
  requests:
    cpu: "500m"
    memory: "512Mi"
  limits:
    cpu: "1"
    memory: "1Gi"
```

Example ingress

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: hrms-ingress
  namespace: hrms-prod-blue
  annotations:
    nginx.ingress.kubernetes.io/proxy-body-size: "20m"
    nginx.ingress.kubernetes.io/ssl-redirect: "true"
spec:
  tls:
    - hosts:
        - hrms.company.internal
      secretName: hrms-tls
  rules:
    - host: hrms.company.internal
      http:
        paths:
          - path: /api
            pathType: Prefix
            backend:
              service:
                name: hrms-api
                port:
                  number: 8000
          - path: /
            pathType: Prefix
            backend:
              service:
                name: hrms-frontend
                port:
                  number: 80
```

Example HPA

```
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: hrms-api-hpa
  namespace: hrms-prod-blue
spec:
  scaleTargetRef:
```

```

apiVersion: apps/v1
kind: Deployment
name: hrms-api
minReplicas: 3
maxReplicas: 10
metrics:
- type: Resource
resource:
  name: cpu
target:
  type: Utilization
  averageUtilization: 65

```

Secret/config item	Storage location	Rotation policy
DATABASE_URL	Secret/Vault	Rotate during maintenance or after incident.
REDIS_URL	Secret/Vault	Rotate quarterly or after personnel change.
OIDC_CLIENT_SECRET	Secret/Vault	Rotate quarterly.
OBJECT_STORAGE_SECRET_KEY	Secret/Vault	Rotate quarterly.
JWT_SIGNING_KEY	Secret/Vault/HSM if available	Rotate with key versioning.
EMAIL_SMTP_PASSWORD	Secret/Vault	Rotate per policy.
FEATURE_FLAGS	ConfigMap	Change-controlled; not secret.
APP_SETTINGS	ConfigMap	Change-controlled.

16. CI/CD, Release Management and Blue-Green Deployment



Figure 10 - CI/CD flow with security gates and blue-green production release.

Pipeline stage	Required gates
Merge request	Code review, branch policy, linked ticket, no secrets in diff.
Static checks	Lint, formatting, type checks, unit tests.
Security scans	SAST, dependency vulnerability scan, container image scan, license check.
Build	Immutable image tag with commit SHA and semantic version.
Database migration test	Run migrations against disposable DB; verify rollback where possible.
Deploy DEV/SIT	Automated deploy and smoke tests.
Deploy UAT	Business UAT and workflow testing.
Production approval	Change ticket, release notes, migration plan, rollback plan.
Blue-green deploy	Deploy inactive color, run smoke/regression, switch ingress/service selector.
Post-deploy monitoring	Error rate, latency, database load, workflow queue, login, critical endpoint checks.

Example CI/CD pipeline skeleton

```

stages:
- test
- security
- build
- deploy

unit_tests:
  stage: test
  script:
    - pip install -r requirements.txt
    - pytest --cov=apps tests/unit

sast_and_dependency_scan:
  stage: security
  script:
    - semgrep --config auto apps/

```

```
- pip-audit -r requirements.txt
```

build_image:

stage: build

script:

```
- docker build -t registry.internal/erp-hrms/api:${CI_COMMIT_SHA} .  
- trivy image registry.internal/erp-hrms/api:${CI_COMMIT_SHA}  
- docker push registry.internal/erp-hrms/api:${CI_COMMIT_SHA}
```

deploy_green:

stage: deploy

when: manual

script:

```
- helm upgrade --install hrms-green infra/helm/hrms \  
  --namespace hrms-prod-green \  
  --set image.tag=${CI_COMMIT_SHA} \  
  --set color=green  
- ./scripts/smoke-test.sh https://hrms-green.company.internal
```

Blue-green step	Instruction
1. Identify inactive color	If blue serves production traffic, deploy to green. If green serves traffic, deploy to blue.
2. Deploy application	Deploy frontend, backend, workers and migrations to inactive color.
3. Run migration compatibility checks	Prefer backward-compatible migrations before traffic switch. Avoid destructive changes in same release.
4. Smoke test	Login, dashboard, workflow submit/approve, job posting, candidate apply, onboarding conversion, attendance correction.
5. Switch traffic	Update ingress/service selector or load balancer target to inactive color.
6. Monitor	Watch error rate, latency, CPU/memory, queue depth, DB locks and audit logs for 30-60 minutes.
7. Rollback if needed	Switch traffic back to previous color. Only rollback DB when explicitly planned and safe.

17. Observability, Logging and Monitoring

Signal	What to collect	Tooling
Metrics	Request rate, error rate, latency, queue depth, DB pool usage, CPU/memory, approval SLA breaches.	Prometheus + Grafana.
Logs	Structured JSON logs with correlation_id, user_id, module, action, object_id and error code.	Loki, ELK or OpenSearch.
Traces	Cross-service traces for API -> DB -> workflow -> event publisher.	OpenTelemetry + Jaeger/Tempo.
Audit	Immutable business audit events, not just technical logs.	audit_log table + secure export/storage.
Synthetic checks	Login, API health, applicant apply, workflow approval, onboarding conversion.	CronJob/blackbox exporter.

Alert	Threshold example	Owner
API high error rate	5xx > 2% for 5 minutes	App/Infra
API latency	p95 > 1.5s for 10 minutes	App/Infra
DB connection exhaustion	Pool usage > 85% for 10 minutes	DBA/Infra
Queue backlog	Pending jobs > 5,000 or oldest job > 15 minutes	App/Infra
Workflow SLA breach	Checker pending > configured SLA	HR Operations
Failed login spike	Failed logins > threshold per user/IP	Security
Audit write failure	Any audit write failure	Security/App - critical
Backup failure	Any failed scheduled backup	DBA/Infra
Object malware scan failure	Any upload scan error	Security/App

Structured log example

```
# Minimum structured log format
{
  "timestamp": "2026-05-23T10:00:00+08:00",
  "level": "INFO",
  "service": "recruitment-service",
  "correlation_id": "corr_REQ-2026-00018",
  "user_id": "USR-00010",
  "action": "JOB_POSTING_SUBMITTED",
  "object_type": "rec_job_posting",
  "object_id": "JP-2026-00033",
  "status": "SUBMITTED",
  "message": "Job posting submitted for checker review"
}
```

18. Backup, Disaster Recovery and Retention

Data store	Backup strategy	Retention
PostgreSQL	Daily full backup + continuous WAL archiving. Restore test monthly.	Daily 30 days, monthly 12 months, yearly as policy requires.
Object storage	Versioning + lifecycle policy + replication/offsite copy.	Match HR document retention policy.
Redis	Usually cache/queue only. Persist only if used for durable queue; otherwise no restore dependency.	Short-term if persistence enabled.
Kafka/RabbitMQ	Topic retention based on event replay requirements.	7-90 days operational; longer if compliance requires.
Secrets/Vault	Encrypted backup of Vault data and unseal keys stored securely.	Per security policy.
Configuration	GitOps repository is source of truth.	Permanent through Git history.

Tier	Target
RPO - Recovery Point Objective	15 minutes for production database if WAL archiving is enabled; otherwise last scheduled backup.
RTO - Recovery Time Objective	4 hours for critical HRMS production services; adjust based on business need.
DR environment	Warm standby for database and object storage if HRMS is business-critical.
Restore testing	At least monthly for database; quarterly full application restore rehearsal.
Backup encryption	Encrypt backups at rest and in transit. Access restricted to DBA/infra with approval.

Example backup CronJob skeleton

```
apiVersion: batch/v1
kind: CronJob
metadata:
  name: postgres-logical-backup
  namespace: hrms-prod-blue
spec:
  schedule: "0 1 * * *"
  jobTemplate:
    spec:
      template:
        spec:
          restartPolicy: OnFailure
          containers:
            - name: backup
```

```

image: postgres:15
command: ["/bin/sh", "-c"]
args:
- |
  pg_dump "$DATABASE_URL" | gzip > /backup/hrms_$(date +%F).sql.gz
  # Upload to secure object storage using approved backup tool/client
envFrom:
- secretRef:
    name: hrms-backup-secret
volumeMounts:
- name: backup-volume
  mountPath: /backup
volumes:
- name: backup-volume
  persistentVolumeClaim:
    claimName: backup-pvc

```

19. Developer Setup and Local Environment

25. Clone the repository and create a feature branch using the team branching standard.
26. Install Docker/Podman, Python/Node tooling, and pre-commit hooks.
27. Copy .env.example to .env.local and fill non-secret local values.
28. Start local dependencies using docker compose.
29. Run database migrations and seed reference data.
30. Create local test users and role mappings.
31. Run backend API, frontend and worker process.
32. Execute unit/integration tests before opening a merge request.

Local environment file example

```

# .env.local example - do not commit real secrets
APP_ENV=local
DEBUG=true
DATABASE_URL=postgres://hrms:hrms@localhost:5432/hrms
REDIS_URL=redis://localhost:6379/0
OBJECT_STORAGE_ENDPOINT=http://localhost:9000
OBJECT_STORAGE_BUCKET=hrms-documents
OIDC_ISSUER=http://localhost:8080/realms/hrms
OIDC_CLIENT_ID=hrms-local
OIDC_CLIENT_SECRET=local-dev-secret
EVENT_BUS_URL=localhost:9092
AUDIT_IMMUTABLE_MODE=false

```

Local docker-compose dependency stack

```

version: "3.9"
services:
  postgres:
    image: postgres:15
    environment:
      POSTGRES_DB: hrms
      POSTGRES_USER: hrms
      POSTGRES_PASSWORD: hrms
    ports: ["5432:5432"]
    volumes: ["pgdata:/var/lib/postgresql/data"]

  redis:
    image: redis:7
    ports: ["6379:6379"]

```

```

minio:
  image: minio/minio
  command: server /data --console-address ":9001"
  environment:
    MINIO_ROOT_USER: minio
    MINIO_ROOT_PASSWORD: minio123
  ports: ["9000:9000", "9001:9001"]

```

```

keycloak:
  image: quay.io/keycloak/keycloak:latest
  command: start-dev
  environment:
    KEYCLOAK_ADMIN: admin
    KEYCLOAK_ADMIN_PASSWORD: admin
  ports: ["8080:8080"]

```

```

volumes:
  pgdata:

```

Seed data category	Examples
Organization	Company, division, department, unit, branch.
Jobs/grades	Job family, job title, grade, pay band, competency profile.
Positions	Approved vacant/occupied positions for sample departments.
Users/roles	Employee, manager, HR maker, HR checker, recruiter, finance checker, talent committee.
Workflow rules	Position approval, job posting approval, offer approval, attendance correction, succession nomination.
Recruitment templates	Job description template, screening questions, interview scorecard, offer template.
Onboarding templates	New hire checklist, IT access, asset request, policy acknowledgement.
Workforce templates	Shift templates, overtime policy, attendance grace period.
Succession templates	Criticality matrix, readiness levels, risk scoring matrix, development plan types.

20. Testing Strategy and Acceptance Criteria

Test type	What to test	Owner
Unit tests	Domain validations, status transitions, scoring rules, permission checks.	Developers
Integration tests	API + database + workflow + outbox + notifications.	Developers/QA
End-to-end tests	Applicant apply to employee conversion; workforce gap to recruitment; succession gap to requisition.	QA/Business
Security tests	Auth, role bypass, IDOR, CSRF, upload malware, audit tampering, privilege escalation.	Security/QA
Performance tests	Applicant portal load, dashboard load, report queries, workflow approvals, attendance imports.	QA/Infra
Disaster recovery tests	Restore DB/object storage and validate application recovery.	Infra/DBA
UAT	Business process acceptance and maker-checker matrix validation.	HRIS/Business

Critical UAT scenario	Expected result
Create approved position and trigger recruitment	Position moves from draft to approved/vacant; requisition can be created only after approval.
Reject job posting	Posting does not go public; maker receives reason and can amend.
Candidate accepts offer	Onboarding case automatically starts; candidate data is locked for controlled conversion.
Complete onboarding and create employee	Employee master draft goes through HR checker; employee assigned

	to approved position.
Attendance correction	Correction request goes to manager/HR checker; audit captures before/after.
Overtime approval	Approved OT flows to payroll export only after attendance matching.
Critical role with no successor	Succession risk increases; strategic recruitment trigger becomes available.
Nominate successor	Manager nomination requires HR checker/Talent Committee approval and evidence.
Maker tries self-approval	System blocks approval and records attempted action.
Unauthorized user accesses confidential succession plan	Access denied and security event logged.

21. Migration, Cutover and Go-Live Runbook

Migration object	Source	Target	Validation
Organization units	HR/Finance master list	core_org_unit	Hierarchy and active status validated.
Cost centers	Finance system	core_cost_center	Cost center ownership and active dates validated.
Jobs/grades	HR job catalog	core_job/core_grade	Grade/pay band mapping validated.
Positions	HR manpower list	core_position	Position count reconciles to approved headcount.
Employees	Legacy HR/payroll	core_person/core_employee/ core_employment	Employee count, manager and department reconciled.
Assignments	Legacy HR	core_employee_assignment	Every active employee has one primary assignment.
Open requisitions	Recruitment tracker/system	rec_job_requisition/ rec_application	Pipeline stage and owner validated.
Succession plans	Confidential HR files	succ_* tables	Access restricted and approved by HR.

33. Freeze legacy master data changes for migration window or define delta capture process.
34. Export source data and store immutable copy in secure migration folder.
35. Cleanse organization, job, grade, position and employee data.
36. Load reference data first: organization, cost center, grade, job and workflow configuration.
37. Load position data and validate headcount by department.
38. Load employee/person/employment/assignment data and reconcile counts.
39. Load recruitment and succession open items if required for go-live.
40. Run automated validation scripts and generate exception report.
41. Business owners sign off migration reconciliation.
42. Cut over integrations, activate users, and enable production access.
43. Monitor first 5 business days with hypercare support.

Go-live checklist item	Status owner
Production namespace deployed and smoke tested.	Infra
TLS certificate valid and monitored.	Infra/Security
SSO role mapping validated.	Security/IAM
Database backup and restore tested.	DBA/Infra
Workflow approval matrix signed off.	HRIS/Business
Seed data and migration reconciliation signed off.	HRIS/DBA
Critical reports validated.	HR/Executive users
Support runbook and escalation contact list published.	Project/Operations
Rollback plan approved.	Change Advisory Board
Hypercare schedule confirmed.	Project team

22. Future-Ready Enhancements

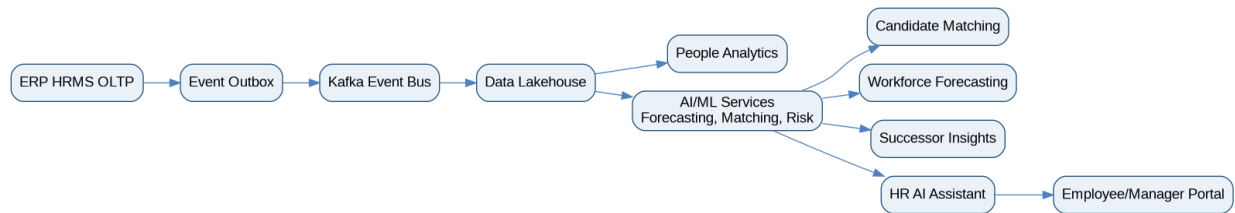


Figure 11 - Future-ready event and AI architecture.

Future capability	Description	Prerequisite
AI candidate matching	Suggest candidate fit based on skills, experience, job requirements and screening answers.	Clean job/skills taxonomy, consent, bias controls.
Workforce forecasting	Predict future headcount gaps, attrition risk, overtime pressure and hiring demand.	Historical workforce, attendance and turnover data.
Successor insights	Summarize strengths, gaps, readiness and development recommendations.	Talent profiles, performance data, learning data and governance.
Skills ontology / skill graph	Standardize skills across jobs, employees, candidates and training.	Skill taxonomy and mapping governance.
HR AI assistant	Natural language access to policies, reports, workflows and employee self-service.	Secure RAG, access filtering and audit.
Mobile-first HR	Attendance, approvals, schedule, applications and onboarding on mobile.	API-first backend and secure mobile authentication.
API ecosystem	Expose controlled APIs for payroll, finance, LMS, IAM and external job boards.	API gateway, service identity and throttling.
Lakehouse analytics	Long-term people analytics and predictive models without stressing OLTP DB.	Event bus, CDC, data catalog and governance.

Future-ready design choice now	Benefit later
Use position_id everywhere	Clean integration between workforce, recruitment and succession.
Use event outbox from v1	Analytics and integrations can be added without rewriting transactions.
Version API and events	Mobile apps and external integrations remain stable.
Capture audit and effective dates	Historical analytics, compliance and dispute handling become easier.
Use configurable workflow rules	New approval policies can be added without code changes.
Normalize skills/job taxonomy	AI matching and succession insights become more accurate.
Separate applicant portal security boundary	Reduces risk from public-facing surface.
Keep reporting read-optimized	Dashboards scale without hurting transaction performance.

23. Implementation Roadmap

Phase	Duration	Deliverables
Phase 0 - Discovery and confirmation	1-2 weeks	Confirm processes, approval matrix, roles, data owners, integration inventory, infrastructure constraints.
Phase 1 - Foundation platform	3-5 weeks	Repo, CI/CD, environments, SSO, database, audit, workflow engine, core HR reference data.
Phase 2 - Recruitment and onboarding	5-8 weeks	Requisition, job posting, applicant portal, candidate pipeline, offer, onboarding handoff and employee conversion.
Phase 3 - Workforce management	6-10 weeks	Position control, workforce plan, assignments, attendance, corrections, overtime, dashboards.
Phase 4 - Succession planning	5-8 weeks	Critical role register, nominations, readiness, risk scoring, development plan and recruitment trigger.
Phase 5 - Integrations and analytics	4-8 weeks	Payroll, IAM, finance, LMS, BI, event bus, reports and audit exports.
Phase 6 - Production hardening	3-6 weeks	Performance, DR, security testing, UAT

		closure, migration, cutover and hypercare.
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Workstream	Key tasks
Business/HRIS	Process signoff, approval matrix, master data rules, UAT scripts, change management.
Development	Backend modules, frontend screens, API contracts, workflow rules, tests, reporting endpoints.
Infrastructure	K8s/OpenShift, database, Redis, object storage, ingress, TLS, DNS, backup, monitoring.
Security	SSO, RBAC, SoD, audit review, threat model, penetration testing, data privacy.
Data migration	Data mapping, cleansing, load scripts, reconciliation, cutover support.
Operations	Runbooks, support model, incident response, monitoring dashboards, hypercare.

24. Appendices and Reference Configuration

24.1 Reference approval matrix

Process	Maker	Checker	Final approver	Notes
Create position	Manager/HR Maker	HR Checker	HR Head + Finance if budget impact	Required before recruitment.
Change position grade	HR Maker	HR Checker	Compensation Head	Pay/grade impact.
Publish job posting	Recruiter	Recruitment Checker	HR Manager	Public visibility.
Issue offer	Recruiter	HR Checker	HR Head + Finance	Budget and compensation impact.
Convert applicant to employee	HR Maker	HR Checker	HR Manager	Creates employee master.
Manual attendance correction	Employee/HR Maker	Manager/HR Checker	Optional HR Manager	Payroll impact.
Overtime	Employee/Manager	Manager + HR Checker	Department Head if threshold exceeded	Payroll cost impact.
Transfer	Manager/HR Maker	HR Checker	Releasing and receiving Dept Head	Org/manager/cost center impact.
Promotion	Manager/HR Maker	HR Checker	HR Head/Compensation	Grade/pay impact.
Mark role critical	HR Talent Maker	HR Talent Checker	Department Head	Succession impact.
Nominate successor	Manager/HR Talent	HR Talent Checker	Talent Committee	Confidential talent decision.

24.2 Sample feature flags

Feature flags

```
FEATURE_RECRUITMENT_PUBLIC_PORTAL=true
FEATURE_ONBOARDING_EMPLOYEE_CONVERSION=true
FEATURE_WORKFORCE_ATTENDANCE_IMPORT=true
FEATURE_SUCCESSION_RISK_SCORING=true
FEATURE_AI_SUCCESOR_INSIGHTS=false
FEATURE_BLUE_GREEN_READINESS=true
FEATURE_PAYROLL_EXPORT=false
FEATURE_STRICT_SOD_ENFORCEMENT=true
```

24.3 Recommended non-functional requirements

Category	Target
Availability	99.5% for MVP; 99.9% for production critical HRMS after HA/DR maturity.
API latency	p95 < 1.5 seconds for common API calls under expected load.
Dashboard latency	p95 < 3 seconds for standard dashboards using read models/materialized views.

Audit durability	No business transaction may be approved if audit write fails.
Upload scanning	All applicant and employee uploads must pass malware scan before being made available.
Data retention	Follow HR/legal policy; configure retention by document and transaction type.
Accessibility	WCAG 2.1 AA target for portals and dashboards.
Localization	Design labels/date/time formats for future localization.
Time zone	Store timestamps in UTC; display in user/org timezone.
Scalability	Stateless app pods; horizontal scaling for frontend/API/workers.

24.4 References reviewed

- [1] SAP HCM product page - describes SAP SuccessFactors HCM as a connected HCM suite supporting core HR/payroll, talent management, HR analytics, workforce planning and employee experience; also describes HRMS capabilities including recruiting, onboarding, time and attendance and succession planning. URL: <https://www.sap.com/products/hcm.html>
- [2] SAP Learning - Employee Central Position Management - describes position management as functionality to create, maintain, staff and activate positions. URL: https://learning.sap.com/courses/sap-successfactors-employee-central-position-management-academy/describing-sap-successfactors-employee-central-position-management_bcd65377-bc96-400d-9df9-c31c228a8ecf
- [3] SAP SuccessFactors successor insights page - describes AI-assisted successor insights based on role alignment, performance, achievements, strengths and development areas. URL: <https://www.sap.com/france/products/hcm/successfactors-succession-development-successor-insights.html>
- [4] SAP Community topic page - workforce management coverage around core HR/time/payroll solution area. URL: <https://pages.community.sap.com/topics/successfactors/core-hr-and-payroll-solutions/workforce-management>
- [5] SAP Help - Integrating SAP SuccessFactors Recruiting Management with Onboarding - describes onboarding as part of the hiring process when integrated with recruiting. URL: <https://help.sap.com/docs/successfactors-onboarding/implementing-onboarding/integrating-sap-successfactors-recruiting-management-with-onboarding>
- [6] SAP SuccessFactors Onboarding product page - describes onboarding, cross-boarding, offboarding and rehire processes through a comprehensive solution. URL: <https://www.sap.com/products/hcm/employee-onboarding.html>

Final implementation reminder

Do not treat this as only a software build. The ERP HRMS will succeed only if the team finalizes HR master data ownership, approval matrices, change management, data migration quality, access governance and operational support before production go-live.